

KAIKOURA, New Zealand

WMO: 936780

Lat: 42.4204S

Lon: 173.6950E

Elev: 108

StdP: 100.04

Time Zone: 12.00 (NZL)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.5	4.2	-4.8	2.6	9.8	-3.3	2.9	9.9	17.1	9.2	15.2	7.8	6.6	280	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.8	24.9	16.1	23.3	15.6	22.0	15.2	17.7	21.8	17.0	21.0	16.4	19.9	3.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.3	11.8	18.6	15.6	11.3	17.9	15.1	10.8	17.4	50.4	22.0	48.2	21.1	46.4	20.1	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.8	13.3	11.4	DB	2.1	29.1	0.8	1.4	1.5	30.1	1.0	31.0	0.5	31.7	-0.1	32.8	(4)
(5)			WB	0.4	19.2	0.8	0.8	-0.1	19.7	-0.6	20.2	-1.1	20.6	-1.7	21.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	16.1	16.2	15.1	12.9	11.3	9.3	8.4	8.9	10.2	11.5	12.8	15.0	(6)	
	DBStd	3.80	2.89	2.62	2.71	2.49	2.67	2.40	2.35	2.45	2.75	2.82	2.88	2.96	(7)	
	HDD10.0	227	0	0	0	4	16	41	62	52	30	16	6	1	(8)	
	HDD18.3	2248	83	71	107	164	219	271	308	293	243	214	166	110	(9)	
	CDD10.0	1063	189	173	159	90	55	21	13	17	37	61	91	157	(10)	
	CDD18.3	41	13	10	7	1	0	0	0	0	0	1	1	8	(11)	
	CDH23.3	137	49	34	18	1	1	0	0	0	0	2	6	27	(12)	
(13)	CDH26.7	10	4	2	2	0	0	0	0	0	0	0	0	2	(13)	
(14) Wind	WSAvg	4.5	4.8	4.5	4.3	4.2	4.3	4.5	4.4	4.4	4.4	4.7	4.8	4.6	(14)	
(15) Precipitation	PrecAvg	948	62	68	89	88	71	82	108	86	71	77	76	68	(15)	
	PrecMax	1378	243	150	191	279	221	326	427	344	198	143	214	237	(16)	
	PrecMin	601	16	23	10	16	0	28	21	2	17	10	23	18	(17)	
	PrecStd	182	46	32	53	55	43	59	90	74	49	34	40	43	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.0	26.7	25.2	22.1	21.3	18.0	17.0	17.8	20.0	22.1	23.6	26.2	(19)	
		MCWB	16.7	17.1	16.2	13.9	12.8	10.4	9.7	10.2	11.4	12.9	14.0	16.1	(20)	
	2%	DB	24.6	24.2	23.1	19.9	18.8	16.1	14.8	15.4	17.8	19.2	21.0	23.6	(21)	
		MCWB	16.1	16.5	15.8	13.3	11.9	9.8	8.9	9.1	10.6	12.2	13.6	15.4	(22)	
	5%	DB	22.9	22.5	21.4	18.3	17.0	14.6	13.6	13.9	16.2	17.6	19.4	21.6	(23)	
		MCWB	15.7	16.2	15.0	12.6	11.2	9.1	8.2	8.4	10.0	11.5	13.3	15.0	(24)	
	10%	DB	21.1	20.7	19.8	16.9	15.4	13.2	12.2	12.6	14.8	16.2	17.9	20.0	(25)	
		MCWB	15.4	15.7	14.4	12.0	10.5	8.6	7.7	7.9	9.3	11.1	12.7	14.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.6	18.7	17.7	15.7	14.4	12.1	11.4	11.3	12.9	14.4	15.8	18.0	(27)	
		MCDB	22.8	22.7	22.5	18.8	18.4	14.9	14.5	15.5	17.9	18.8	19.7	22.5	(28)	
	2%	WB	17.6	17.7	16.7	14.6	13.1	11.1	10.1	10.1	11.7	13.2	14.9	16.8	(29)	
		MCDB	21.6	21.7	20.8	17.6	15.9	13.6	12.6	13.0	15.8	17.3	18.8	21.0	(30)	
	5%	WB	16.8	17.0	16.0	13.7	12.3	10.3	9.2	9.4	10.8	12.4	14.2	16.1	(31)	
		MCDB	20.8	20.5	19.5	16.4	15.1	12.8	11.7	11.9	14.4	16.3	17.8	19.9	(32)	
	10%	WB	16.2	16.4	15.3	13.0	11.5	9.6	8.5	8.8	10.1	11.7	13.4	15.4	(33)	
		MCDB	19.6	19.4	18.3	15.4	14.2	12.0	11.0	11.1	13.3	15.0	16.8	18.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	5.8	5.6	5.1	4.7	4.5	4.3	4.5	5.4	6.0	6.1	6.0	(35)	
		MCDBR	9.4	9.1	8.8	7.7	7.6	6.9	6.4	6.8	7.9	8.7	9.1	9.1	(36)	
	5% WB	MCWBR	4.3	3.9	4.1	3.9	4.0	3.8	3.9	4.0	4.1	4.5	4.6	4.0	(37)	
		MCWBR	7.5	7.4	7.0	6.1	5.7	5.2	5.0	5.2	6.7	7.5	7.7	7.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.331	0.323	0.309	0.297	0.286	0.280	0.275	0.288	0.301	0.315	0.316	0.323	(40)	
		taud	2.596	2.640	2.675	2.687	2.671	2.651	2.678	2.665	2.602	2.587	2.613	2.631	(41)	
	Ebn at Noon		996	978	950	894	836	805	841	891	941	975	1005	1009	(42)	
		Edh at Noon	101	92	81	68	58	54	57	68	85	96	99	99	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.48	5.75	4.49	2.98	2.02	1.52	1.76	2.55	3.76	5.15	6.30	6.46	(44)	
	RadStd		0.52	0.54	0.24	0.25	0.12	0.09	0.12	0.17	0.21	0.32	0.49	0.46	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83	N/A
(47) Regional (6 neighbors)		N/A	N/A	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page